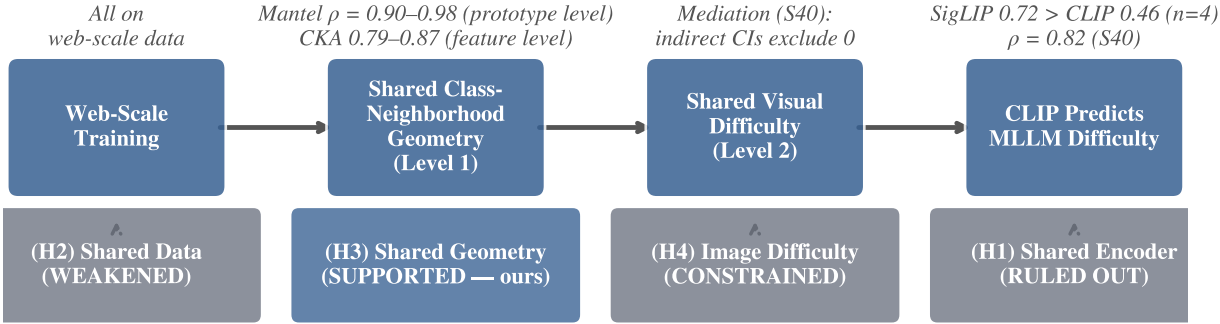
All on web-scale data
Mantel ρ = 0.90–0.98 (prototype level)
CKA 0.79–0.87 (feature level)
Mediation (S40): indirect CIs exclude 0
SigLIP 0.72 > CLIP 0.46 (n=4)
ρ = 0.82 (S40)
Web-Scale Training
Shared Class-Neighborhood Geometry (Level 1)
Shared Visual Difficulty (Level 2)
CLIP Predicts MLLM Difficulty
(H2) Shared Data (WEAKENED)
(H3) Shared Geometry (SUPPORTED — ours)
(H4) Image Difficulty (CONSTRAINED)
(H1) Shared Encoder (RULED OUT)